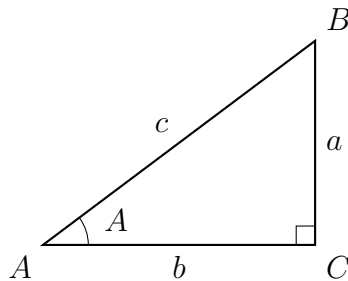


Right Triangles: Which Method?

choosing between the Pythagorean theorem, special right triangles and trigonometry

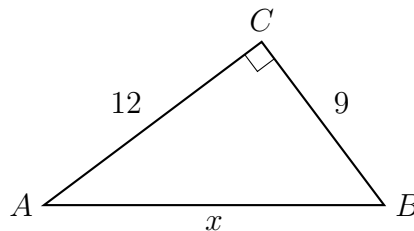
Directions

- Decide first *which tool the givens allow*, then compute. Three sides and no angle is Pythagoras; a 45° or a $30^\circ/60^\circ$ angle gives an exact special-triangle ratio; any other angle needs sine, cosine or tangent.
- Round lengths to two decimal places and angles to two decimal places. Write the unit on every answer.
- Each diagram is drawn to scale from that problem's own givens.



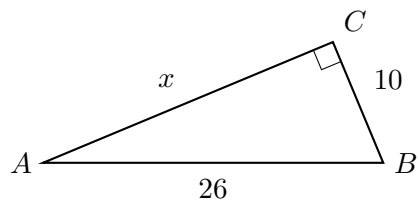
How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. The two legs of a right triangle measure 9 cm and 12 cm. Find the hypotenuse, in centimetres.



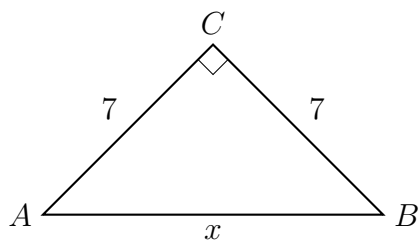
Answer: _____ cm

2. A right triangle has hypotenuse 26 in and one leg 10 in. Find the other leg, in inches.



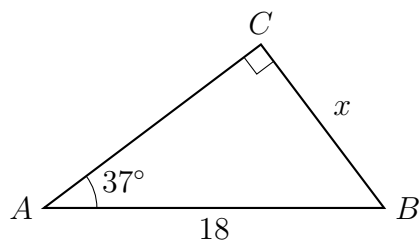
Answer: _____ in

3. An isosceles right triangle has both legs equal to 7 cm. Find the hypotenuse, in centimetres.



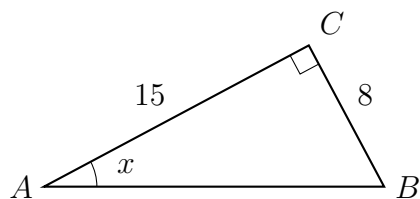
Answer: _____ cm

4. A right triangle has hypotenuse 18 m and one acute angle 37° . Find the side opposite that angle, in metres.



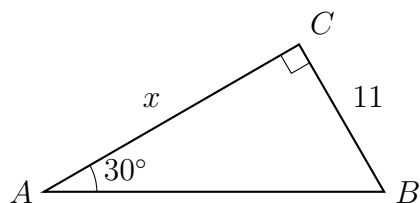
Answer: _____ m

5. A right triangle has legs of 8 and 15. Find the measure of the acute angle opposite the leg of length 8, in degrees.



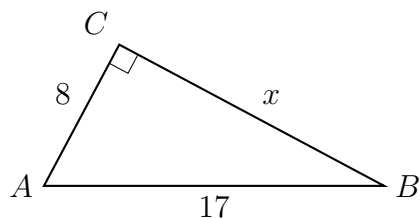
Answer: _____ degrees

6. In a right triangle one acute angle is 30° and the leg opposite it measures 11 ft. Find the other leg, in feet.



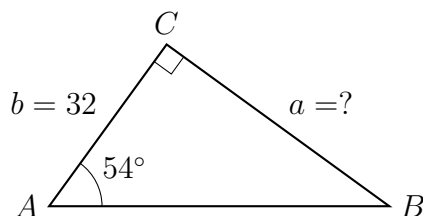
Answer: _____ ft

7. A 17 ft ladder rests against a vertical wall with its foot 8 ft from the base of the wall. Find how far up the wall the ladder reaches, in feet.



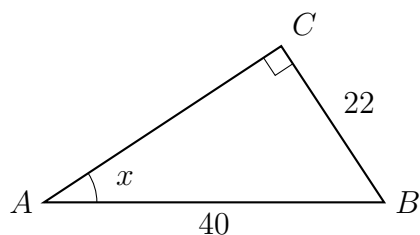
Answer: _____ ft

8. A path runs 32 m from a gate straight to the foot of a vertical tower. From the gate the tower top is seen at 54° above the horizontal. Find the height of the tower, in metres.



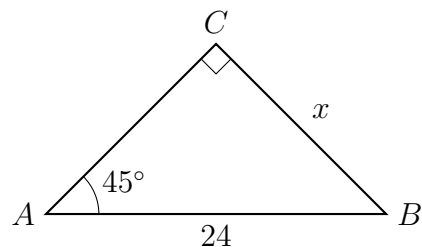
Answer: _____ m

9. A support cable 40 ft long runs from the ground to a point 22 ft up a mast. Find the angle the cable makes with the ground, in degrees.



Answer: _____ degrees

10. A square tile is cut along a diagonal of length 24 cm, giving two identical right triangles.



- (a) Find the length of each leg of one of those triangles, in centimetres.

Answer: _____ cm

- (b) Explain how the special right-triangle relationship gives the same answer, and why it is the quicker route here.